ROGUE ACCESS POINT DETECTION REPORT

Generated: 2026-04-16 12:28:33 | **Total frames:** 24

24	8	2	3
Total Frames	Unique BSSIDs	Rogue APs	Suspicious APs

Finding Severity Breakdown

Severity	Count	Risk Level
HIGH	8	Critical — immediate action required
MEDIUM	2	Notable — investigate promptly
LOW	0	Minor — monitor for escalation

Access Point Risk Summary

BSSID	SSID(s)	Score	Findings	Verdict
de:ad:be:ef:ca:fe	NETWORK_2, NETWORK_1	14	5	ROGUE
00:00:00:00:00:06	NETWORK_2	5	2	ROGUE
00:00:00:00:00:03	NETWORK_1	3	1	SUSPICIOUS
00:00:00:00:00:12	NETWORK_5	3	1	SUSPICIOUS
00:00:00:00:00:14	NETWORK_1	3	1	SUSPICIOUS

Detailed Findings

SSID Spoofing — HIGH

SSID 'NETWORK_1' is broadcast by 6 distinct BSSIDs: ['00:00:00:00:00:03', '00:00:00:00:00:04', '00:00:00:00:00:06', 'de:ad:be:ef:ca:fe', '00:00:00:00:00:02', '00:00:00:00:00:14']

SSID Spoofing — HIGH

SSID 'NETWORK_2' is broadcast by 3 distinct BSSIDs: ['00:00:00:00:00:06', '00:00:00:00:00:04', 'de:ad:be:ef:ca:fe']

MAC Inconsistency	HIGH
Frame transmitter (de:ad:be:ef:ca:fe) ≠ BSSID (de:ad:be:ef:ca:ff) for SSID 'NETWORK_1' at t=1682624219.478	

MAC Inconsistency	HIGH
Frame transmitter (de:ad:be:ef:ca:fe) ≠ BSSID (de:ad:be:ef:ca:ff) for SSID 'NETWORK_2' at t=1682625745.310	

Probe Response Flood	MEDIUM
BSSID 00:00:00:00:00:06 sent 4 ProbeResp frames (threshold=3); SSIDs: ['NETWORK_2', 'NETWORK_1']	

Probe Response Flood	MEDIUM
BSSID de:ad:be:ef:ca:fe sent 6 ProbeResp frames (threshold=3); SSIDs: ['NETWORK_1', 'NETWORK_3', 'NETWORK_2']	

Unknown BSSID	HIGH
BSSID de:ad:be:ef:ca:fe ('NETWORK_1') is not in the authorised whitelist	

Unknown BSSID	HIGH
BSSID 00:00:00:00:00:12 ('NETWORK_5') is not in the authorised whitelist	

Unknown BSSID	HIGH
BSSID 00:00:00:00:00:14 ('NETWORK_1') is not in the authorised whitelist	

Detection Methodology

This system implements signature-based analysis of IEEE 802.11 wireless frames captured in a simulated campus environment. Six detection rules are applied to identify rogue access points based on frame anomalies, SSID conflicts, and signal behaviour patterns.

Rule	Severity	Description
SSID Spoofing	HIGH	Detects when the same SSID is advertised by more than one distinct BSSID across captures. In a legiti
MAC Inconsistency	HIGH	In Beacon and ProbeResponse frames the transmitter address (addr2) must equal the BSSID field (add
Unknown BSSID	HIGH	When an authorised BSSID whitelist is provided, any BSSID not appearing in that list is flagged immedia
RSSI Anomaly	MEDIUM	Computes the Z-score of each BSSID's RSSI values across all frames. Readings deviating beyond 2.5 s
Probe Response Flood	MEDIUM	A rogue AP acting as a honeypot responds eagerly to every ProbeRequest. Any source BSSID sending
MAC Timestamp Drift	LOW	The MAC-layer timestamp in Beacon frames should advance at the same rate as the system wall clock.

Risk Scoring System

Severity	Points	Verdict Thresholds
HIGH	3 pts	ROGUE (HIGH CONFIDENCE): Score ≥ 5

MEDIUM	2 pts	SUSPICIOUS: Score ≥ 3
LOW	1 pt	LOW RISK: Score < 3

Recommendations

#	Priority	Action
1	HIGH	Immediately isolate and investigate all ROGUE-flagged BSSIDs. Verify physical presence of the device on campus network infras
2	MEDIUM	Monitor SUSPICIOUS-flagged BSSIDs closely. Cross-reference against the authorised AP whitelist maintained by network admin
3	HIGH	Investigate all HIGH-severity SSID Spoofing and MAC Inconsistency findings — these are strong indicators of evil twin or frame ir
4	LOW	Maintain and regularly update the authorised BSSID whitelist. Schedule periodic wireless scans at minimum every 24 hours.
5	LOW	Educate campus users to verify network authenticity before connecting and to avoid networks with unfamiliar or duplicate SSIDs.